



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

2004
YEAR 11 HALF-YEARLY EXAMINATION

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – One and a half hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.

Total Marks – 72

- Attempt all questions.
- All questions are of equal value.
- Each section is to be answered in a separate booklet, labeled Section A (Questions 1, 2), Section B (Questions 3, 4) and so on.

Examiner: *A.M. Gainford*

Note: This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section A**Marks**
6**Question 1**

- (a) Simplify $x(x - y) - y(y - x)$.
- (b) Evaluate $\sqrt{\frac{3^2 + 12^2}{231 - 12^2}}$ correct to three significant figures.
- (c) Express in simplest reduced form: $\left(\frac{2}{5}\right)^{10} \times \left(\frac{15}{4}\right)^{10} \times \left(\frac{2}{3}\right)^9$

Question 2**6**

- (a) Express $0.\overline{72}$ as a common fraction in lowest terms.
- (b) Simplify $2\sqrt{\frac{9}{4}} + \sqrt[3]{\frac{8}{125}}$ completely.
- (c) State the value of the following, in exact terms:
- (i) $\tan 60^\circ$
- (ii) $\cos \frac{11\pi}{6}$

Section B

(Start a new booklet)

Question 3**6**

- (a) Solve for x : $x(x - 4) = 5$
- (b) Express $\frac{1}{\sqrt{3} - 2}$ with rational denominator.
- (c) Find the value of x if $\sqrt{x} = \sqrt{50} - \sqrt{18}$.

Question 4**6**

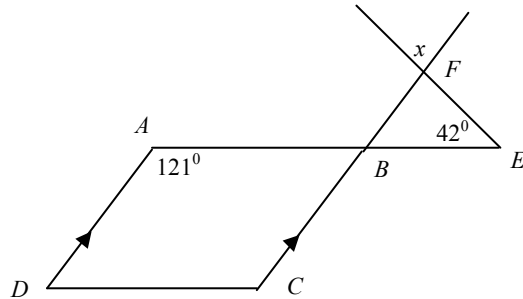
- (a) Expand and simplify: $\sqrt{(a-4)(a+4)+16}$, $a > 0$.
- (b) Three legs of a triangular sailing course have lengths 8 km, 10 km, and 16 km.
- (i) Draw a sketch showing this information.
- (ii) Calculate the size of the smallest angle, correct to the nearest minute.
- (c) On a number line sketch the solution of $4 - x \geq 3$.

Section C
(Start a new booklet.)

Question 5**6**

- (a) In the diagram $AD \parallel BC$.

Copy the diagram onto your worksheet and find the value of x , giving reasons.



- (b) Sketch the graph of $y = 2\sin(90^\circ - x)$ in the domain $-90^\circ \leq x \leq 360^\circ$.

Question 6**6**

- (a) Find the shortest distance between the parallel lines $y = 2x$ and $4x - 2y + 7 = 0$.
- (b) Factorise completely each of the following:
- (i) $5x^2 - 9xy - 2y^2$
- (ii) $x^2 + 4x + 4 - y^2$
- (c) Solve $|5 - 2x| = 3$.

Section D
(Start a new booklet.)

Question 7

6

- (a) Solve the following system of simultaneous equations:

$$\begin{aligned}2(x - y) &= -7 \\ 6x + 8y &= 35\end{aligned}$$

- (b) Draw a neat sketch of the graph of the function $y = \sqrt{9 - x}$.

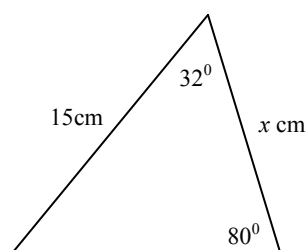
State the domain and range of this function.

- (c) Show that $f(x) = \frac{4^x + 4^{-x}}{2}$ is an even function.

Question 8

6

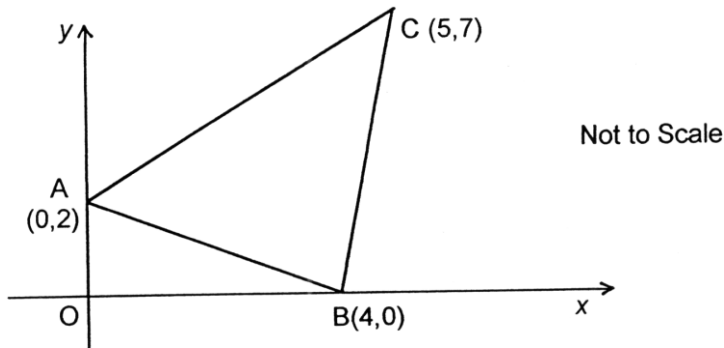
- (a) Find the least value of the quadratic expression $x^2 - 4x + 10$, and state the x -value at which it occurs.
- (b) Find the values of k for which the quadratic equation $3x^2 + 2x + k = 0$ has no real roots.
- (c) Find the value of x correct to two decimal places.



Section E
(Start a new booklet.)

Question 9

6



The diagram shows the points $A(0, 2)$, $B(4, 0)$, and $C(5, 7)$.

Copy the diagram onto your work sheet.

- (a) Find the co-ordinates of M , the mid-point of AB .
- (b) Show that the gradient of AB is $-\frac{1}{2}$.
- (c) Find the equation of the perpendicular bisector of AB .
- (d) Show that the perpendicular bisector of AB passes through C .
- (e) What type of triangle is ABC ? (Give a reason for your answer.)

Question 10

6

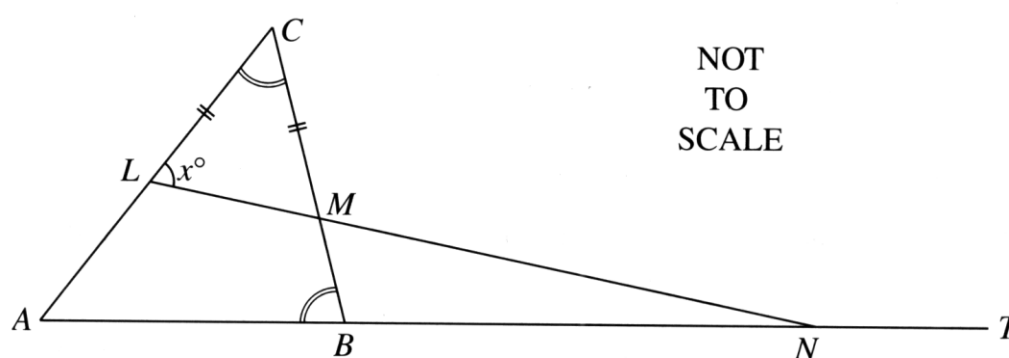
- (a) Sketch on the number plane the region in which all three of the following inequalities are satisfied:

$$x - y + 2 > 0$$

$$2x + y \geq -2$$

$$x - 3 \leq 0$$

- (b)



In the diagram ABC is an isosceles triangle with $\angle ABC = \angle ACB$. The line LMN is drawn as shown so that $CL = CM$, and $\angle CLM = x^\circ$.

Copy or trace the diagram to your booklet.

- (i) Show that $\angle ABC = (180 - 2x)^\circ$.
- (ii) Hence show that $\angle TNL = 3x^\circ$.

This is the end of the paper.



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Sample Solutions

Section	Marker
A	Mr Fuller
B	Mr Dowdell
C	Mr Boros
D	Ms Nesbitt
E	Ms Opferkuch

Section A
Question 1

(a) $x(x-y) - y(y-x)$

$$= x^2 - xy - y^2 + xy$$

$$= x^2 - y^2$$

(b) $\sqrt{\frac{3^2 + 12^2}{231 - 12^2}}$

$$= \sqrt{\frac{9 + 144}{231 - 144}}$$

$$= \sqrt{\frac{153}{87}}$$

$$= 1.33$$

(c) $\left(\frac{2}{5}\right)^{10} \times \left(\frac{15}{4}\right)^{10} \times \left(\frac{2}{3}\right)^9$

$$= \left(\frac{3}{2}\right)^{10} \times \left(\frac{2}{3}\right)^9$$

$$= \left(\frac{3}{2}\right)^9 \times \left(\frac{2}{3}\right)^9 \times \left(\frac{3}{2}\right)$$

$$= \frac{3}{2}$$

Question 2

(a) let $x = 0.727272\ldots$

$$100x = 72.727272\ldots$$

$$99x = 72$$

$$x = \frac{72}{99}$$

$$x = \frac{8}{11}$$

$$(b) \quad 2\sqrt{\frac{9}{4}} + 3\sqrt{\frac{8}{125}}$$

$$= 2 \times \frac{3}{2} + \frac{2}{5}$$

$$= 3 + \frac{2}{5}$$

$$= 3\frac{2}{5}$$

$$(c) \quad (i) \quad \tan 60^\circ = \sqrt{3}$$

$$(ii) \quad \cos \frac{11\pi}{6} = \cos \frac{\pi}{6}$$

$$= \frac{\sqrt{3}}{2}$$

Q3 (a) $x(x-4) = 5$

$$x^2 - 4x - 5 = 0$$

$$(x-5)(x+1) = 0$$

$$x = 5 \text{ or } x = -1 \quad (2)$$

(b) $\frac{1}{\sqrt{3}-2} \times \frac{\sqrt{3}+2}{\sqrt{3}+2}$

$$= \frac{\sqrt{3}+2}{3-4}$$

$$= -(\sqrt{3}+2) \quad (2)$$

(c) $\sqrt{x} = \sqrt{50} - \sqrt{18}$

$$= 5\sqrt{2} - 3\sqrt{2}$$

$$= 2\sqrt{2}$$

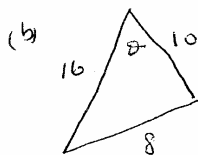
$$\therefore x = 8 \quad (2)$$

4 (a) $\sqrt{(a-4)(a+4)+16}$

$$= \sqrt{a^2 - 16 + 16}$$

$$= \sqrt{a^2}$$

$$= a \quad (\text{as } a > 0) \quad (2)$$



$$\cos \theta = \frac{16^2 + 10^2 - 8^2}{2 \times 16 \times 10}$$

$$= \frac{292}{320}$$

$$= \frac{73}{80}$$

$$\therefore \theta = 24.1468 \dots$$

$$\approx 24^\circ 9' \quad (2)$$

(c) $4 - x \geq 3$

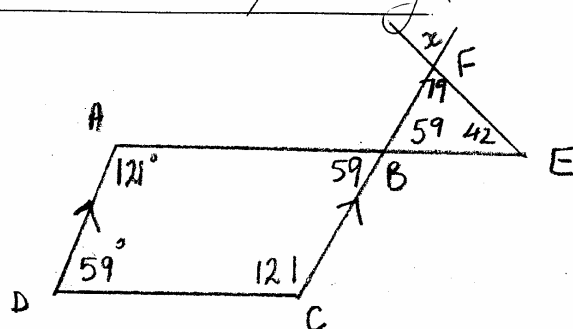
$$-x \geq -1$$

$$x \leq 1$$



(2)

5 (a)



(1)

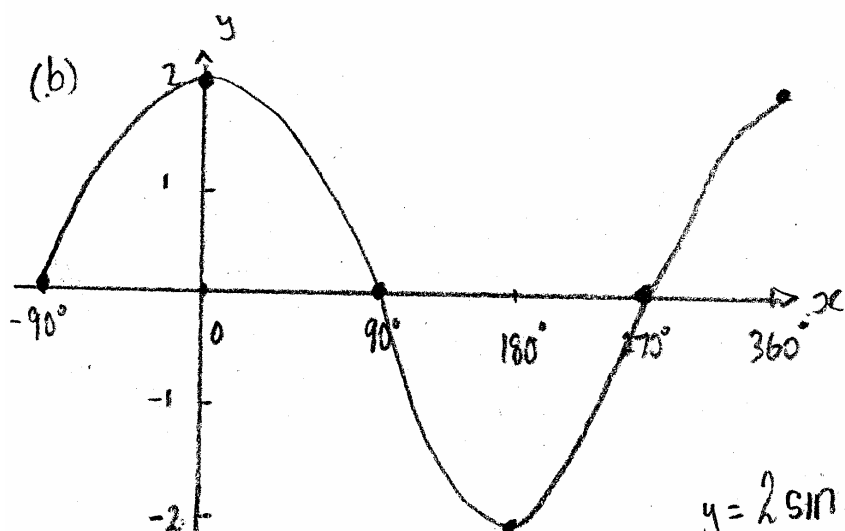
$$\hat{ABC} = 180 - 121 = 59^\circ \text{ co interior angles}$$

$$\hat{ABC} = \hat{FBE} = 59^\circ \text{ vert. opp.}$$

$$\hat{BFE} = 180 - (59 + 42) = 79$$

$$\therefore x = \hat{BFE} = 79^\circ \text{ vert. opp.}$$

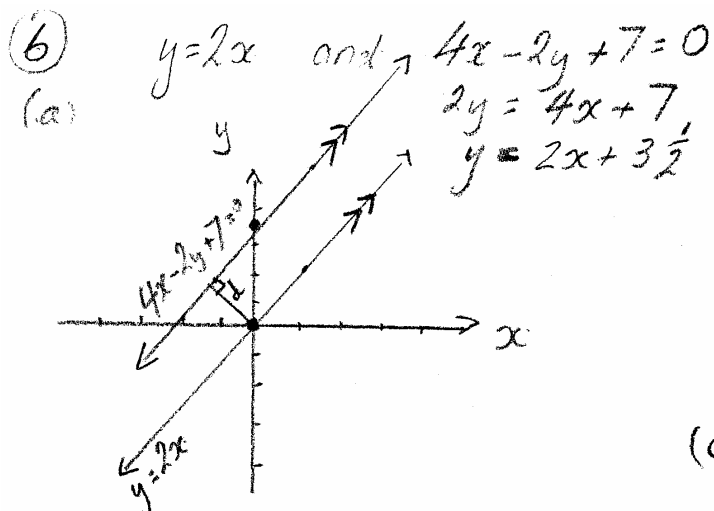
(2)



(3)

$$y = 2 \sin(90 - x) = 2 \cos x$$

Since $\sin(90 - x) = \cos x$



Use $(0,0)$ on $4x - 2y + 7 = 0$

$$d = \left| \frac{7}{\sqrt{16+4}} \right|$$

$$= \frac{7}{\sqrt{20}} = \frac{7}{2\sqrt{5}} \text{ or } (\div 1.56)$$

$$\frac{7}{2\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{7\sqrt{5}}{10} \text{ units } \textcircled{2}$$

(b) (i) $5x^2 - 9xy - 2y^2$

$$= (5x + y)(x - 2y) \textcircled{1}$$

(ii) $x^2 + 4x + 4 - y^2$

$$= (x+2)^2 - y^2$$

$$= (x+2-y)(x+2+y) \textcircled{2}$$

(c) $|5 - 2x| = 3$

$$\begin{array}{l} \swarrow \quad \searrow \\ 5 - 2x = 3 \quad 5 - 2x = -3 \\ -2x = -2 \quad -2x = -8 \\ x = 1 \quad x = 4 \end{array}$$

①

Q7(a)

$$2x - 2y = -7 \quad (1)$$

$$6x + 8y = 35 \quad (2)$$

$$8x - 8y = -28 \quad (1 \times 4)$$

$$14x = 7$$

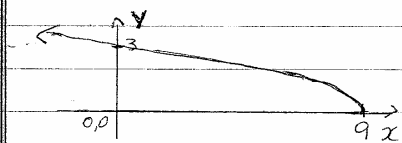
$$x = \frac{1}{2}$$

$$2x \cdot \frac{1}{2} - 2y = -7 \quad (1)$$

$$-2y = -8$$

$$y = 4 \quad x = \frac{1}{2} \quad (3)$$

(b)



Domain: $x \leq 9$

Range: $y \geq 0$ (2)

(c) $f(-x) = \frac{4^{-x} + 4^x}{2} = f(x)$

$\therefore f(x)$ is an even function

(1)

Q8(a)

least value at centre axis

$$x = -\frac{b}{2a}$$

$$x = 2$$

least value = 6

or

$$x^2 - 4x + 10 = (x - 2)^2 + 6$$

vertex (2, 6)

least value = 6 when $x = 2$ (1)

(b) No real root when $D < 0$

$$b^2 - 4ac < 0$$

$$2^2 - 4 \times 3 \times k < 0$$

$$\therefore 12k < -4$$

$$k > \frac{1}{3} \quad (2)$$

(c) Third angle = $180 - 80 - 32$

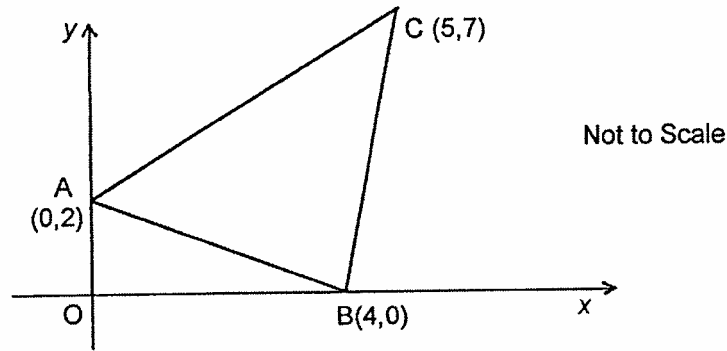
$$= 68^\circ$$

$$\frac{x}{\sin 68} = \frac{15}{\sin 80}$$

$$x = 15 \frac{\sin 68}{\sin 80}$$

$$x = 14.12 \text{ (2dp)} \quad (3)$$

Question 9



$$(a) \quad M: x = \frac{4+0}{2}, y = \frac{2+0}{2} \quad (1)$$

$$= 2 \quad = 1$$

$$\therefore M(2,1)$$

$$(b) \quad m_{AB} = \frac{0-2}{4-0} \quad (1)$$

$$= -\frac{1}{2}$$

$$\therefore m_{AB} = -\frac{1}{2}$$

$$(c) \quad m_1 \times m_2 = -1 \quad (1)$$

$$-\frac{1}{2} \times m_2 = -1$$

$$m_2 = 2$$

Using $y - y_1 = m(x - x_1)$, where $x = 2, y = 1, m_2 = 2$

$$y - 1 = 2(x - 2)$$

$$\therefore 2x - y - 3 = 0$$

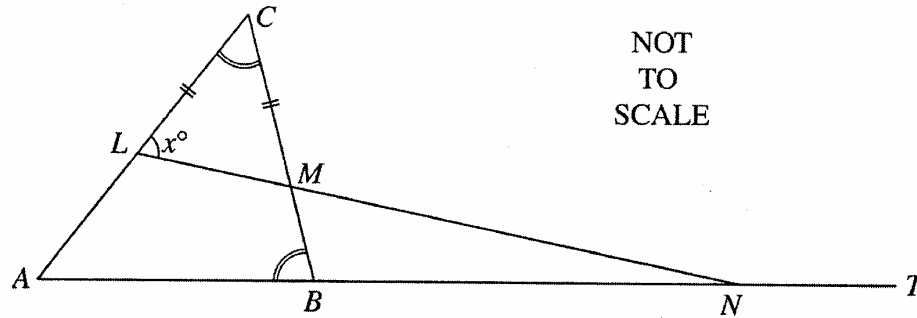
$$(d) \quad \text{Using } 2x - y - 3 = 0, \text{ where } x = 5, y = 7 \quad (1)$$

$$2 \times 5 - 7 - 3 = 0$$

$$\therefore LHS = RHS$$

- (e) In $\triangle ABC$ (2)
- $$d_{AC} = d_{BC} = \sqrt{50}$$
- $$d_{AB} = \sqrt{20}$$
- $\therefore \triangle ABC$ is isosceles

Question 10



- (b) (i) In $\triangle CLM$ (1)
- $$\angle CLM = x^\circ \text{ (data)}$$
- $$\angle CML = x^\circ \text{ (base } \angle, \text{ isos. } \triangle)$$
- $$\therefore \angle ACB = 180 - 2x \text{ (}\angle \text{ sum of } \triangle)$$
- In $\triangle ABC$
- $$\angle ACB = \angle ABC \text{ (data)}$$
- $$\therefore \angle ABC = 180 - 2x$$
- (ii) In $\triangle MBN$ (2)
- $$\angle BMN = \angle CML \text{ (vert. opp. } \angle \text{ s)}$$
- $$= x^\circ$$
- $$\angle MBN = 180 - \angle ABC \text{ (supp. } \angle \text{ s)}$$
- $$= 180 - (180 - 2x)$$
- $$= 2x$$
- $$\therefore \angle TNL = \angle BMN + \angle MBN \text{ (ext. } \angle = \text{sum of two opp. int. } \angle \text{ s)}$$
- $$= x + 2x$$
- $$= 3x$$

(a)

